Nobel Prep

SAT Practice Exam

Mock Test 1

Math — Module 1

22 Questions · 35 Minutes · No Calculator Required for All

Source	June 2023 Form B
Section	Math
Module	1 of 2
Time Allowed	35 minutes
Questions	22
Student Name	
Date	

DIRECTIONS

The questions in this section address a number of important math skills. Use of a calculator is permitted for all questions. A reference sheet, calculator, and these directions are always available.

Multiple-choice questions — Choose the best answer. The correct answer is one of the choices given.

Student-produced response questions — Enter your answer in the box. Answers may be positive or negative numbers, decimals, or fractions. Do not enter symbols such as a percent sign, comma, or dollar sign.

DIRECTIONS

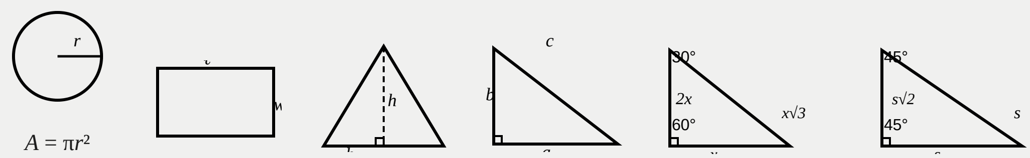
The questions in this section address a number of important math skills. Use of a calculator is permitted for all questions.

NOTES

Unless otherwise indicated:

- All variables and expressions represent real numbers.
- Figures provided are drawn to scale.
- All figures lie in a plane.
- The domain of a given function *f* is the set of all real numbers *x* for which *f*(*x*) is a real number.

REFERENCE



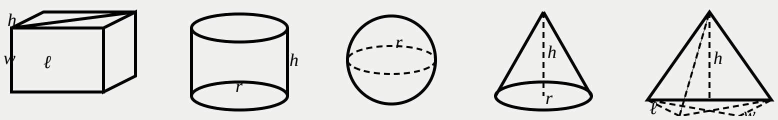
$A = \pi r^2$
$C = 2\pi r$

$A = \ell w$

$A = \frac{1}{2}bh$

$c^2 = a^2 + b^2$

Special Right Triangles



$V = \ell wh$

$V = \pi r^2 h$

$V = \frac{4}{3}\pi r^3$

$V = \frac{1}{3}\pi r^2 h$

$V = \frac{1}{3}\ell wh$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π.

The sum of the measures in degrees of the angles of a triangle is 180.

CONTINUE ▶

DIRECTIONS

The questions in this section address a number of important math skills. Use of a calculator is permitted for all questions. For multiple-choice questions, choose the best answer. For student-produced response questions, enter your answer in the box provided.

UV Index Data

UV index	Number of days
1	9
2	15
3	12
4	13

1 Mark for Review

The table summarizes the UV index value recorded by a research assistant at noon each day for 49 days. According to the table, a UV index value of 1 was recorded on how many days?

(A) 4
(B) 9
(C) 40
(D) 49

2 Mark for Review

To make a bookcase, a woodworker charged a one-time fee plus \$17 per hour worked. The equation $17h + 45 = 164$ represents this situation, where h is the number of hours worked. Which of the following is the best interpretation of 164 in this context?

(A) The one-time fee, in dollars
(B) The number of hours worked
(C) The charge per hour, in dollars
(D) The total charge, in dollars

3

Mark for Review

In December 2017, the lowest temperature recorded in a certain city was 40 degrees Fahrenheit (°F) and the highest temperature recorded was 90°F. Which inequality is true for all values of t, where t represents any temperature, in °F, recorded in the city in December 2017?

A $40 \le t \le 90$

B $t \le 90$

C $t \le 50$

D $t \ge 90$

4

Mark for Review

The function f is defined by $f(x) = 6(2x + 4)$. For what value of x does $f(x) = 48$?

A 2

B 6

C 8

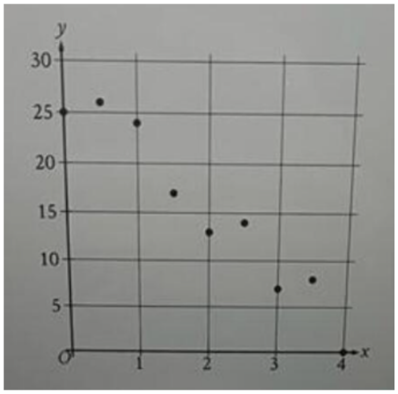
D 22

5 Mark for Review

The ratio of green tiles to blue tiles in a piece of artwork is 5 to 2. If there are 16 blue tiles in the piece of artwork, how many green tiles are there?

Answer

The scatterplot shows the relationship between two variables, x and y.



6 Mark for Review

Which of the following equations is the most appropriate linear model for the data shown?

(A) $y = -6 + 28x$

(B) $y = 6 - 28x$

(C) $y = 28 + 6x$

(D) $y = 28 - 6x$

CONTINUE

7

Mark for Review

A scientist analyzed a soil sample with a mass of 900 grams and determined that it contained 189 grams of water. What is the percentage of water, by mass, in this soil sample?

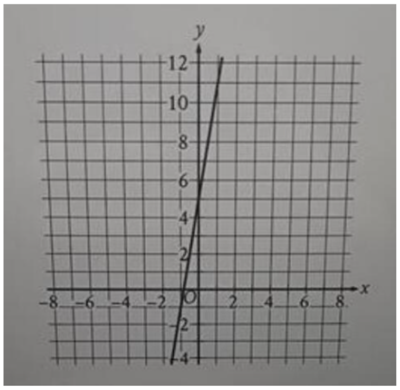
Ⓐ 9%

Ⓑ 9.9%

Ⓒ 18.9%

Ⓓ 21%

Line *n* is shown in the *xy*-plane.



8

Mark for Review

Line *n* is shown in the *xy*-plane. Line *k* (not shown) is perpendicular to line *n*. What is the slope of line *k*?

Ⓐ $-\frac{1}{5}$

Ⓑ $-\frac{1}{6}$

Ⓒ 5

Ⓓ 6

9

Mark for Review

$mx + ky = -83$
$2x + ky = 22$

In the given system of equations, m and k are constants. The graphs of these equations in the xy-plane intersect at the point $(5, y)$. What is the value of m?

Answer

10

Mark for Review

For the linear function f, the graph of $y = f(x)$ in the xy-plane passes through the points $(0, 2)$ and $(3, 3)$. What is the slope of $y = f(x)$?

Answer

11

Mark for Review

A certain book has 250 pages, and 21 of these pages have an illustration. If one of the book's pages is selected at random, what is the probability of selecting a page with an illustration? (Express your answer as a decimal or fraction, not as a percent.)

Answer

12

Mark for Review

$b^2 + 5c = 9d$

The given equation relates the real numbers b, c, and d, where $d > \frac{5}{9}c$. Which equation correctly expresses b in terms of c and d?

A) $b = \frac{9d + 5c}{2}$

B) $b = \frac{9d - 5c}{2}$

C) $b = \pm\sqrt{9d + 5c}$

D) $b = \pm\sqrt{9d - 5c}$

13 Mark for Review

A right circular cylinder has a height of 4 meters (m) and a base with a radius of 18 m. What is the volume, in m^3, of the cylinder?

(A) 4π

(B) 22π

(C) 288π

(D) 1296π

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14 Mark for Review

What is the y-intercept of the graph of $3x + 2y = 96$ in the xy-plane?

(A) (0, 5)

(B) (0, 6)

(C) (0, 32)

(D) (0, 48)

15

Mark for Review

If $\frac{6}{7}p + 12 = 54$, what is the value of $7p$?

Answer

16

Mark for Review

A certain neighborhood had a population of 1340 in 2009. Each year for the next 5 years, the population of the neighborhood increased by approximately 3% of the population the previous year. Which of the following equations represents the population, N, of the neighborhood t years after 2009, where $t \leq 5$?

A) $N = 0.03(1340)^t$

B) $N = 1340(0.3)^t$

C) $N = 1340(1.03)^t$

D) $N = 1.03(1340)^t$

CONTINUE

17

Mark for Review

In the xy-plane, which of the following does NOT contain any points (x, y) that are solutions to $7x + 4y > 12$?

(A) The region where $x > 0$ and $y > 0$

(B) The region where $x < 0$ and $y > 0$

(C) The region where $x < 0$ and $y < 0$

(D) The region where $x > 0$ and $y < 0$

18

Mark for Review

In triangle RST, the measure of angle R is 10 degrees, and the measure of angle T is 50 degrees. Point L lies on RS, point K lies on ST, and LK is parallel to RT. What is the measure, in degrees, of angle SKL?

Answer

19 Mark for Review

An auditorium has seats for 3200 people. Tickets to attend a show at the auditorium currently cost \$8.00. For each \$1.00 increase to the ticket price, 100 fewer tickets will be sold. This situation can be modeled by the equation $y = -100x^2 + 2400x + 25600$, where x represents the increase in ticket price, in dollars, and y represents the revenue, in dollars, from ticket sales. If this equation is graphed in the xy-plane, at what value of x is the maximum of the graph?

(A) 8

(B) 12

(C) 24

(D) 32

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20 Mark for Review

$(x + 3)^2 + (y - 4)^2 = 25$

In the xy-plane, the graph of the given equation is a circle. Which point lies on this circle?

(A) $(-3, 4)$

(B) $(3, -4)$

(C) $(\sqrt{11} + 3, \sqrt{14} - 4)$

(D) $(\sqrt{11} - 3, \sqrt{14} + 4)$

21 Mark for Review

The expression $\dfrac{x^{20}(x-4)}{5x^2} + \dfrac{4x^{20}}{5x^2}$ is equivalent to $\dfrac{1}{5x^c}$, where c is a constant and $x > 0$. What is the value of c?

(A) 4

(B) 5

(C) 19

(D) 21

22 Mark for Review

Triangle ABC is similar to triangle DEF, where angle A corresponds to angle D, and angles C and F are right angles. The length of AB is 2.4 times the length of DE. If $\tan A = \dfrac{21}{20}$, what is the value of $\sin D$?

Answer

Answer Key

Mock Test 1 · Math Module 1 (22 Questions) · June 2023 Form B

1. **B** 9 days
2. **D** Total charge
3. **A** $40 \le t \le 90$
4. **A** $x = 2$
5. **40** SPR
6. **D** $y = 28 - 6x$
7. **D** 21%
8. **A** $-1/5$
9. **−19** SPR
10. **1/3** SPR
11. **21/250** SPR
12. **D** $b = \pm\sqrt{(9d-5c)}$
13. **D** 1296π
14. **D** (0, 48)
15. **343** SPR (7p = 343)
16. **C** $N = 1340(1.03)^t$
17. **C** $x<0$, $y<0$
18. **130** SPR (degrees)
19. **B** $x = 12$
20. **D** $(\sqrt{11}-3, \sqrt{14}+4)$
21. **C** $c = 19$
22. **21/29** SPR

* SPR = Student-Produced Response (숫자 직접 입력)

* Q20 선택지 C,D는 무리수 포함 좌표 ($\sqrt{11}\pm3$, $\sqrt{14}\pm4$ 형태)